

**B.Sc (HONS.) IN CSE FIRST YEAR PART-I, FIRST SEMESTER
EXAMINATION, 2021**

STRUCTURED PROGRAMMING LANGUAGE

[According to the New Syllabus]

Subject Code : 510201

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four questions.]

- | | Marks |
|---|-------|
| 1. (a) What is structured programming? Explain different control structures used in structured programming. | 5 |
| (b) Define pseudo-code. Write down the differences between algorithm and flowchart. | 1+4=5 |
| (c) What do you mean by preprocessor directive? Why and when do we use the #include and #define directives? | 1+4=5 |
| (d) Write an algorithm to determine whether a number is odd or even. | 5 |
| 2. (a) With example describe the rules for ++ and -- operator. | 4 |
| (b) What is symbolic constant? Write down the rules for declaring symbolic constant. | 5 |
| (c) Find output of the following statements :
int x = -10, y = 3 ;
printf("%d\n", x/y * 10) ;
y = -y ;
printf("%d", x/y * 10) ; | 3 |
| (d) Write a C program to verify whether a given year is a leap year or not. | 5 |
| (e) What do you understand by overflow and underflow of data? | 3 |
| 3. (a) What is loop? Write down the differences between entry control and exit control loop. | 1+4=5 |
| (b) Describe the following statements with general form and flowchart :
(i) nested if else
(ii) while | 5 |
| (c) When do you need to use switch statement? Mention the differences between break and continue statements. | 5 |
| (d) Write a C program to input an integer number and display it in reverse order. | 5 |

[Please turn over

	Marks
4. (a) What is string? Briefly explain string handling functions with example.	6
(b) Write a C program to copy one string into another and count the number of characters copied.	5
(c) Define linear array. How to declare and initialize a linear array?	5
(d) What is pointer? What are some arithmetic operators that are permitted to use with pointer? Give example.	4
5. (a) How can a function be defined? Describe. Also mention some merits of user defined function.	7
(b) Why prototyping is essential in a program?	3
(c) What do you understand by call by value and call by reference?	4
(d) Define structure and union. Write down the difference in between array and structure.	6
6. (a) What do you mean by file and file mode? Describe different types of file mode used in C.	2+4=6
(b) Distinguish the following :	6
(i) printf() and fprintf() function	
(ii) scanf() and fscanf() function	
(c) What is recursion? Write down the differences between recursion and iteration.	1+2=3
(d) Write a C program to read from a file and display it on the screen:	5

B.Sc. (HONS.) IN CSE FIRST YEAR, FIRST SEMESTER EXAMINATION, 2021

CALCULUS

[According to the New Syllabus]

Subject Code : CSE-510205

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four questions.]

1. (a) Define domain and range with example.

Marks

(b) Define continuity and differentiability of the given function. A function $f(x)$ is defined as follows :

4

12

$$f(x) = \begin{cases} 3 + 2x & \text{when } -\frac{3}{2} \leq x < 0 \\ 3 - 2x & \text{when } 0 \leq x < \frac{3}{2} \\ -3 - 2x & \text{when } x \geq \frac{3}{2} \end{cases}$$

Test the continuity and differentiability at $x = 0$ and $x = \frac{3}{2}$.

(c) By (δ, ϵ) definition, prove that $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = 6$.

4

2. (a) Find $\frac{dy}{dx}$ (any three) :

$$f(x) = \frac{x^2 - 9}{x - 3}$$

4 × 3 = 12

(i) $y = x^x + \frac{1}{x^x}$

(ii) $y = \tan^{-1} \left(\frac{1 - \cos x}{1 + \cos x} \right)^{\frac{1}{2}}$

(iii) $xy = \sin(x + y)$

(iv) Differentiate $x^{\sin^{-1} x}$ with respect to $\sin^{-1} x$.

(b) State and prove Leibnitz's theorem.

8

3. (a) State and prove Rolle's theorem.

8

(b) Discuss the application of Rolle's theorem to the function $f(x) = x^2$ in $(-1, 1)$.

6

(c) Find the maximum and minimum value of the function

6

$f(x) = x^5 - 5x^4 + 5x^3 - 1$.

[Please turn over]

Marks

4×5=20

4. Evaluate (any five) :

(i) $\int \frac{7x-9}{x^2-2x+35} dx$

(ii) $\int \frac{dx}{5+4\cos x}$

(iii) $\int \frac{dx}{(1+x^2)\sqrt{1-x^2}}$

(iv) $\int \frac{1-\sin x}{1+\sin x} dx$

(v) $\int \frac{dx}{\sqrt{x}+\sqrt{x+1}}$

(vi) $\int \frac{2x+1}{x^2+x-2} dx$

$\frac{1}{2} \ln \left| \frac{1-\cos 2x}{1+\cos 2x} \right| + \frac{1}{2} \ln \left| \frac{1-\sin 2x}{1+\sin 2x} \right|$

5. (a) Evaluate :

6

$$\lim_{n \rightarrow \infty} \left[\frac{1}{n} + \frac{1}{\sqrt{n^2-1^2}} + \frac{1}{\sqrt{n^2-2^2}} + \dots + \frac{1}{\sqrt{n^2-(n-1)^2}} \right]$$

(b) Prove the $\beta(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$

8

(c) Evaluate (any two) :

3×2=6

(i) $\int_0^{\infty} \frac{(\tan^{-1} x)^2}{1+x^2} dx$

(ii) $\int_0^a \sqrt{\frac{a+x}{a-x}} dx$

(iii) $\int_0^{\frac{\pi}{2}} \cos^8 x dx$

6. (a) Find the area between the parabola $y^2 = 4x$ and the straight line $y = 8x-1$

6

(b) Find the parameter of the circle $x^2 + y^2 = a^2$

7

(c) Find the volume of the solid generated by revolving the cardioid $r = a(1-\cos\theta)$ about initial line.

7

B.Sc. (HONS.) IN CSE FIRST YEAR, FIRST SEMESTER EXAMINATION, 2021

PHYSICS

[According to the New Syllabus]

Subject Code : CSE-510207

Time—3 hours

Full marks—80

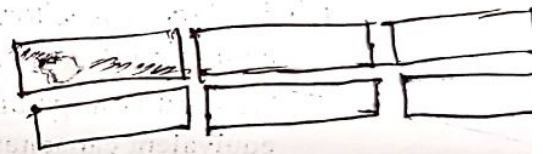
[N.B. The figures in the right margin indicate full marks. Answer any four questions.]

1. (a) Define electric field and electric field strength. Marks 4
 (b) Define electric lines of force. What are the properties of electric lines of force. 5
 (c) What is capacitance? Derive an expression for the energy stored in a charged capacitor. 6
 (d) If three capacitors with capacitances 4, 3 and $2\mu\text{F}$ are so connected that the first and second are in series combination and the third is in parallel combination with them, then what is their equivalent capacitance? 5
2. (a) State Gauss's law. Deduce Coulomb's law from Gauss's law. 7
 (b) It is given that a section of an infinite line of charge has linear charge density λ . Calculate the electric field strength E at a distance Y from the line. 5
 (c) Calculate the potential due to a point charge. 5
 (d) What is the electric potential at the surface of a gold nucleus? 3
 The radius is $6.6 \times 10^{-15}\text{m}$ and the atomic number $Z = 79$.
3. (a) State Biot-Savart's law. Also express it in its vector form. 1+7 5
 (b) Define magnetic flux. With necessary diagram derive an expression for force acting between two parallel conductors. 1+7=8
 (c) Calculate the amount of torque produced on a current carrying loop placed in a magnetic field. 7
4. (a) Define the terms current, current density, resistance, resistivity and conductivity. 5
 (b) State and explain Ohm's law. 4
 (c) Draw and explain basic operation of the Wheatstone bridge. 6
 (d) Draw RC circuit and explain transient response of this circuit. 5

[Please turn over]

Marks

5. (a) Distinguish between self-inductance and mutual-inductance. 4
- (b) Derive the expression for the energy stored in magnetic field. 6
- (c) Deduce the expression for inductance in series when two coils are in series aiding connection. 5
- (d) What is coefficient of coupling? Derive the expression of coupling coefficient. 1+4=5
6. (a) State and explain Faraday's law of electromagnetic induction. 5
- (b) Distinguish between Diamagnetic, Paramagnetic and Ferromagnetic materials. 4
- (c) Distinguish among Dia, Para and Ferromagnetic substances. 6
- (d) Describe hysteresis curve of magnetization. 5



B.Sc (HONS.) IN CSE FIRST YEAR FIRST SEMESTER

EXAMINATION, 2021

ELECTRICAL AND ELECTRONIC CIRCUIT

[According to the New Syllabus]

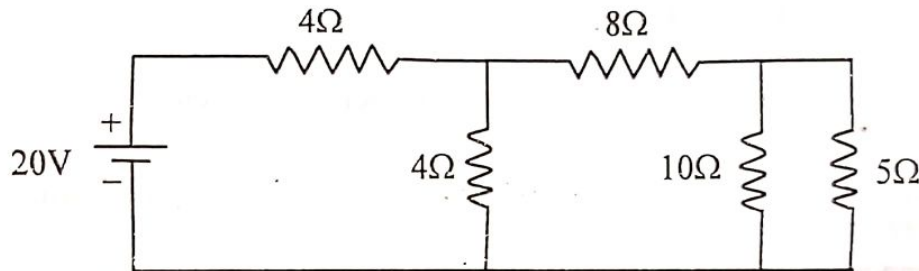
Subject Code : CSE-510203

Time—3 hours

Full marks—80

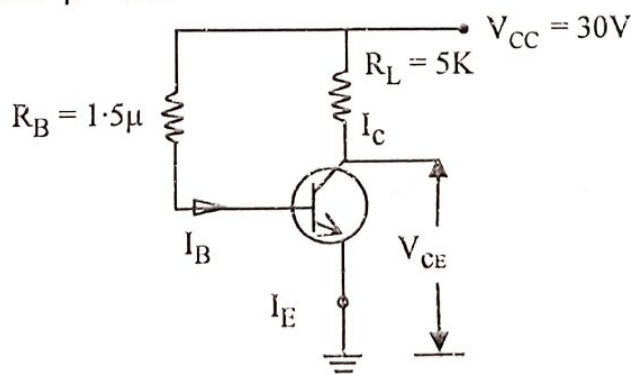
[N.B. The figures in the right margin indicate full marks. Answer any four questions.]

- | | Marks |
|---|-------|
| 1. (a) Define RMS value and Form Factor. | 4 |
| (b) Find the impedance of a series RLC circuit. | 4 |
| (c) Discuss the basic principle of AC generator. | 4 |
| (d) State Norton's theorem. Calculate the current through the 5Ω resistor of the following circuit using Norton's theorem. | 2+6=8 |



- | | |
|--|---|
| 2. (a) Define : Conductor and semiconductor. | 4 |
| (b) Describe how P-type and N-type materials are formed. | 6 |
| (c) What is P-N junction? Describe I—V characteristics of a forward biased P-N junction. | 6 |
| (d) Define : Clipper and Clamper circuit. | 4 |
| 3. (a) Define dc load line and Q-point of a transistor circuit. | 6 |
| (b) For the Common Emitter Circuit (CE) shown in figure. Draw the dc load line and mark the Q-point on it. | 6 |

Assume $\beta = 100$



- | | |
|---|---|
| (c) Draw a Common-Base (CB) transistor circuit and explain its operation. | 5 |
| (d) Find the relation between α and β of a transistor. | 3 |

[Please turn over]

B.Sc (HONS.) IN CSE FIRST YEAR, FIRST SEMESTER EXAMINATION, 2021

ENGLISH

[According to the New Syllabus]

Subject Code : 510209

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four questions. Each part of a question must be answered sequentially.]

1. ✓ Read the following passage and answer the questions below :

Beauty is easy to appreciate but difficult to define. As we look around, we discover beauty in pleasurable objects and sights in nature, in the laughter of children, in the kindness of strangers. But asked to define, we run into difficulties. Does beauty have an independent objective identity? Is it universal or dependent on our sense perceptions? Does it lie in the eye of the beholder?—We ask ourselves. A further difficulty arises when beauty manifests itself not only by its presence, but by its absence as well as when we are repulsed by ugliness and desire beauty. But then ugliness has as much a place in our lives as beauty, or may be more as when there is widespread hunger and injustice in a society. Philosophers have told us that beauty is an important part of life, but isn't ugliness a part of life too? And if art has beauty as an important ingredient, can it confine itself only to a projection of beauty? Can art ignore what is not beautiful, assume a tolerable proportion as something that represents some truth about life. John Keats, the romantic poet, wrote in his celebrated "Ode on a Grecian Urn" Beauty is truth, truth is beauty, by which he means that truth, even if it's not pleasant, becomes beautiful at a higher level.

Marks

Write answers to the following questions :

- | | |
|--|---|
| (a) (i) "Beauty is truth, truth is beauty."—Do you agree or disagree with this statement? Give reasons. | 5 |
| (ii) When do we run into difficulties? | |
| (b) What is the main idea and what are the supporting ideas of the passage? | 5 |
| (c) Write down the meaning of the following words in English and make your own sentences with them (any five):
Confine, universal, manifest, ignore, tolerable, define, desire. | 5 |
| (d) Write a summary of the passage. | 5 |

[Please turn over]

✓ 2. (a) Frame WH-Questions from the following sentences :

5

- (i) Everybody wishes to be happy.
- (ii) I saw him in the meeting.
- (iii) Tania has purchased five books.
- (iv) The train leaves the stations at 7 p.m.
- (v) Rana went home to enjoy Eid.

(b) Use the right form of verbs in the following sentences :

5

- (i) Recently they (join) the team.
- (ii) His choice of words (be) excellent.
- (iii) Why you (not come) home yesterday?
- (iv) If I had a camera, I (take) a snap.
- (v) It is high time we (take) the decision for the development of the project.

(c) Correct the following sentences :

5

- (i) Every morning I am going for a walk.
- (ii) Would you mind to sing a song?
- (iii) He resembles to his father.
- (iv) We congratulated him for his success.
- (v) No one should indulge to smoke.

obediently

(d) Combine each of the following groups of sentences into one sentence :

5

- (i) There was a gentleman. He lived in a village. He had three sons. They always quarrelled with one another.
- (ii) You may like it, you may not like it. I shall give you good advice. It is my duty to do so.

3. (a) Amplify the following :

Rome was not built in a day.

10

(b) Translate into English :

বাংলাদেশ একটি প্রাকৃতিক দুর্যোগের দেশ। প্রায় প্রতি বছর এখানে বন্যা হয়। ঘূর্ণিঝড়ে হাজার হাজার মানুষ মারা যায়। আর ক্ষতি হয় কোটি কোটি টাকার ফসল ও সম্পদ। প্রাকৃতিক দুর্যোগ আমাদের সকল প্রচেষ্টাকে ব্যর্থ করে দেয়। তবুও আমাদের সকল বাধা উপেক্ষা করে এগিয়ে যেতে হয়।

10

- | | Marks |
|---|-------|
| 4. Write an <u>essay</u> on any one of the followings: | 20 |
| (a) Fifty years of Independence | |
| (b) Climate change and its effects | |
| (c) Uses and Abuses of Internet. | |
| 5. (a) Write a dialogue between two friends about a cricket match. | 10 |
| (b) Write a report for the press on the seminar "Save the rivers, save the country". | 10 |
| 6. (a) Write a paragraph on any one of the followings : | 10 |
| (i) Global Warming | |
| (ii) Food Adulteration. | |
| (b) Write an application to the Principal for permission to arrange Inter-Department friendly football match. | 10 |